

Assignment-03

Trends and cloud Services overview.

1. Benefits of Serverless Architecture

Serverless architecture allows developers to build and run applications without managing infrastructure.

Benefit Include:

- Scalability: Automatically Scales with demand.
- Cost-Efficiency: Pay only for actual usage, reducing costs.
- Reduced Operational overhead: No need to manage servers, allowing developers to focus on code.
- Faster Time-to-Market: Accelerates development and deployment cycles.

2. Progressive web APPS (PWAs)

PWAs are web application that offer native app-like experiences. Key features:

- Offline Access: Functionality without an internet connection.
- Installability: Can be added to the user's home screen.
- Responsive Design: Adapts to various screen size and devices.
- Push Notification: Engage users with timely update.

3. Role of AI and Machine Learning in Software Architecture

AI and ML are transforming software architecture by:

- Enhancing User Experience: Personalized recommendations and intelligent interfaces.
- Automating processes: predictive analytics and decision-making support.
- Improving security: Anomaly detection and threat prediction.
- Optimizing Performance: Resource allocation and system tuning based on usage pattern.

Cloud Computing Service Models

- Infrastructure as a Service (IaaS):
Provides virtualized computing resources over the internet.
 - use case: Flexible and scalable hosting solution.
- Platform as a Service (PaaS):
Offers a platform allowing customers to develop, run and manage applications.
 - use case: Streamlined application development without managing underlying infrastructure.